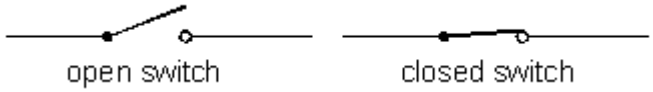
Q1. A pupil makes some electrical circuits.
Each circuit contains a motor and a buzzer.
The motor and the buzzer can be switched on or off by three switches, A, B and C.

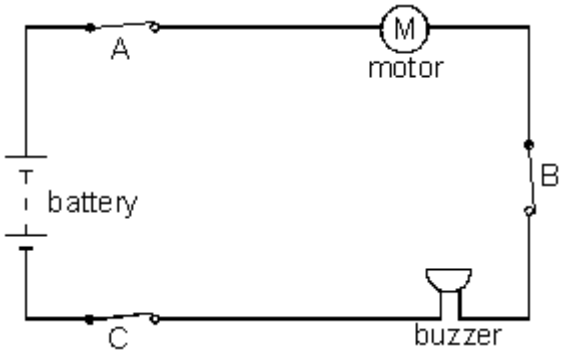
Symbols for an open switch and a closed switch are shown below.



By each circuit diagram, state whether:

the motor is **on** or **off**;
the buzzer is **on** or **off**.

(i)

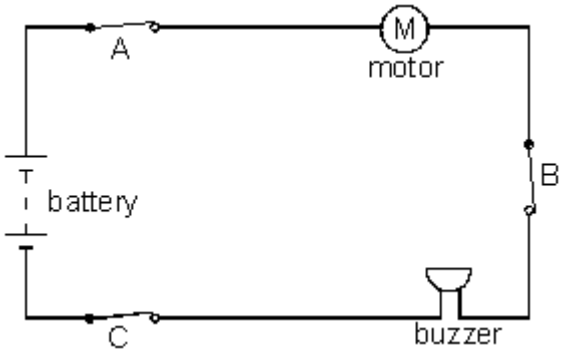


The motor is .....

The buzzer is .....

1 mark

(ii)

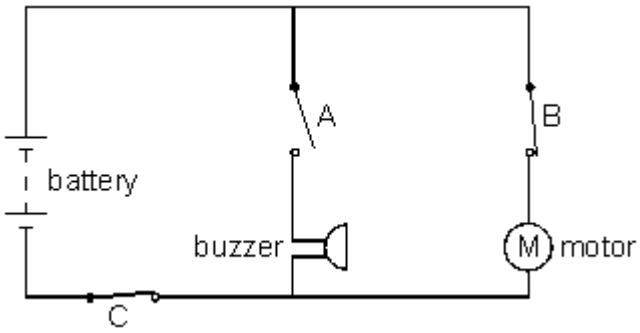


The motor is .....

The buzzer is .....

1 mark

(iii)

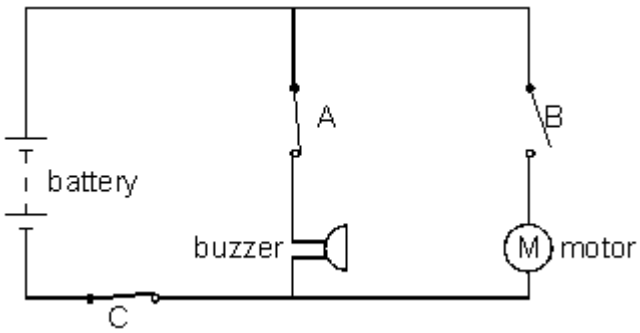


The motor is .....

The buzzer is .....

1 mark

(iv)

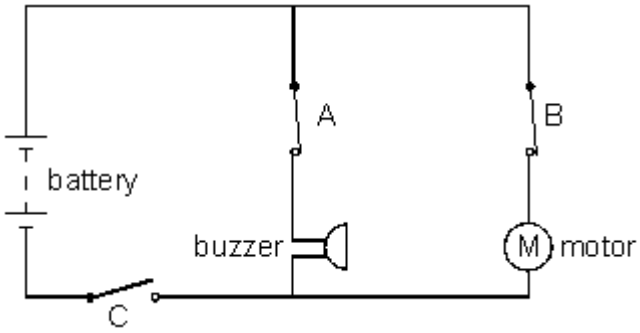


The motor is .....

The buzzer is .....

1 mark

(v)



The motor is .....

The buzzer is .....

1 mark

Maximum 5 marks

Mark schemes

Q1.

(i) off
off

both *answers are required for the mark*

1 (L4)

(ii) on
on

both *answers are required for the mark*

1 (L4)

(iii) on
off

both *answers are required for the mark*
answers must be in the correct order

1 (L4)

(iv) off
on

both *answers are required for the mark*
answers must be in the correct order

1 (L4)

(v) off
off

both *answers are required for the mark*

1 (L4)

[5]